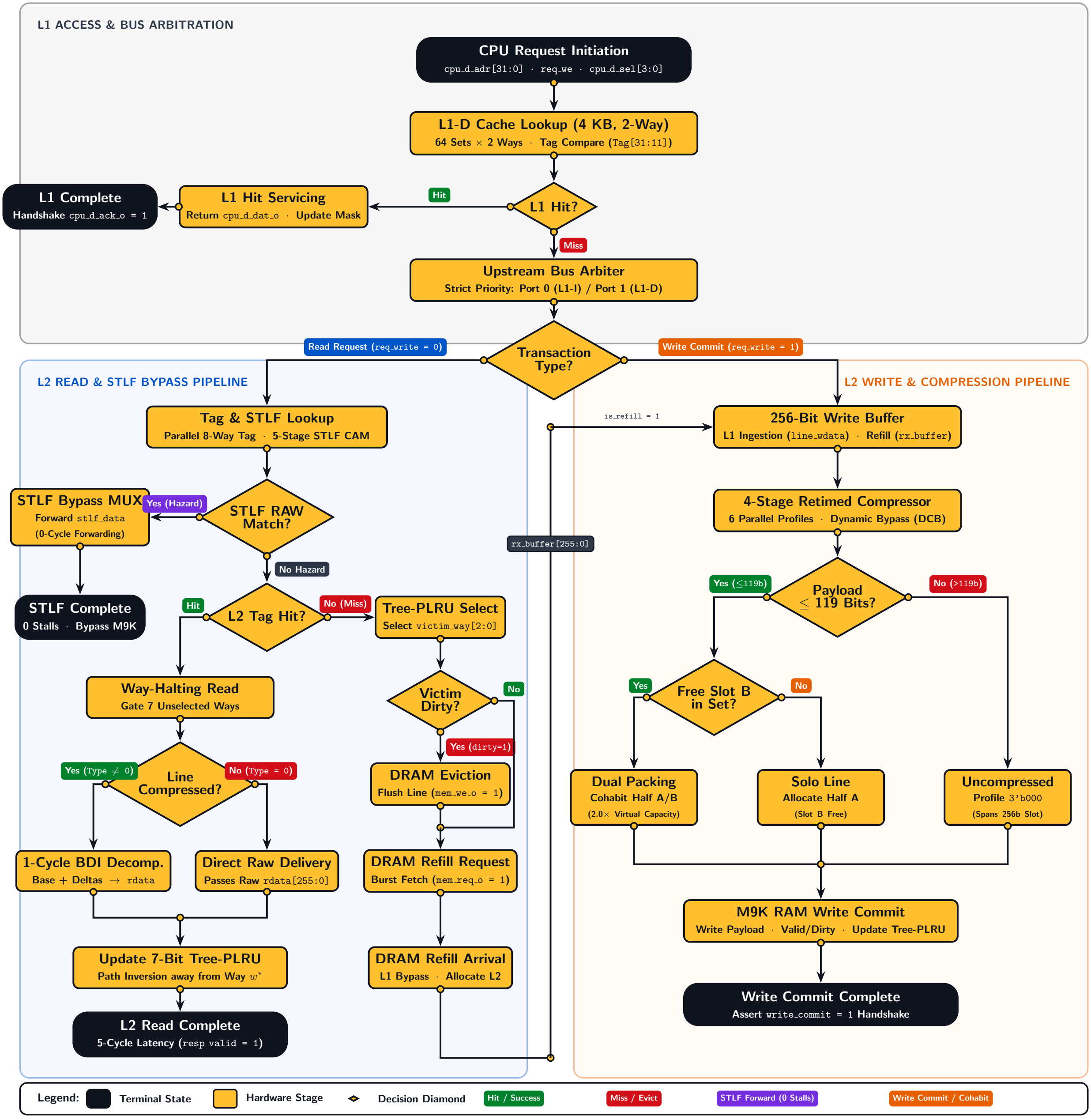

L1 ACCESS & BUS ARBITRATION
CPU Request Initiation
cpu_d_adr[31:0] · req_we · cpu_d_sel[3:0]
L1-D Cache Lookup (4 KB, 2-Way)
64 Sets × 2 Ways · Tag Compare (Tag[31:11])
L1 Hit?
Hit
L1 Hit Servicing
Return cpu_d_dat_o · Update Mask
L1 Complete
Handshake cpu_d_ack_o = 1
Miss
Upstream Bus Arbiter
Strict Priority: Port 0 (L1-I) / Port 1 (L1-D)
Transaction Type?
Read Request (req_write = 0)
Write Commit (req_write = 1)
L2 READ & STLF BYPASS PIPELINE
Tag & STLF Lookup
Parallel 8-Way Tag · 5-Stage STLF CAM
STLF RAW Match?
Yes (Hazard)
STLF Bypass MUX
Forward stlf_data
(0-Cycle Forwarding)
STLF Complete
0 Stalls · Bypass M9K
No Hazard
L2 Tag Hit?
Hit
No (Miss)
Tree-PLRU Select
Select victim_way[2:0]
Victim Dirty?
No
Yes (dirty=1)
DRAM Eviction
Flush Line (mem_we_o = 1)
DRAM Refill Request
Burst Fetch (mem_req_o = 1)
DRAM Refill Arrival
L1 Bypass · Allocate L2
Way-Halting Read
Gate 7 Unselected Ways
Line Compressed?
Yes (Type ≠ 0)
No (Type = 0)
1-Cycle BDI Decomp.
Base + Deltas → rdata
Direct Raw Delivery
Passes Raw rdata[255:0]
Update 7-Bit Tree-PLRU
Path Inversion away from Way w^*
L2 Read Complete
5-Cycle Latency (resp_valid = 1)
is_refill = 1
rx_buffer[255:0]
L2 WRITE & COMPRESSION PIPELINE
256-Bit Write Buffer
L1 Ingestion (line_wdata) · Refill (rx_buffer)
4-Stage Retimed Compressor
6 Parallel Profiles · Dynamic Bypass (DCB)
Payload ≤ 119 Bits?
Yes (≤119b)
No (>119b)
Free Slot B in Set?
Yes
No
Dual Packing
Cohabit Half A/B
(2.0× Virtual Capacity)
Solo Line
Allocate Half A
(Slot B Free)
Uncompressed
Profile 3'b000
(Spans 256b Slot)
M9K RAM Write Commit
Write Payload · Valid/Dirty · Update Tree-PLRU
Write Commit Complete
Assert write_commit = 1 Handshake
Legend:
Terminal State
Hardware Stage
Decision Diamond
Hit / Success
Miss / Evict
STLF Forward (0 Stalls)
Write Commit / Cohabit